

ARYAN SOOD

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Education

Indian Institute of Technology Roorkee

CGPA: 8.175/10 (3.538/4)*

Bachelor of Technology in Electrical Engineering

Aug. 2024 – 2028

Experience

Lexsi Labs

August 2026 – Present

AI Research Intern

- Contributing to the CuratorKit library, developing methods for long-context data generation thereby improving scalability.
- Specifically improving reliability for sensitive synthetic workloads which require private and accurate long documents.

Auric AI Labs

March 2026 – May 2026

Computer Vision Intern

- Generated 2D top-view renders from 3D military-object meshes using **BlenderProc**, forming a synthetic EO data pipeline for object detection.
- Built a semantic suggestion pipeline by adapting *Seeing the Unseen* for EO data and adding deterministic placement guardrails for satellite imagery.
- Trained a **RemoteCLIP-UNet** with frozen CLIP encoders and a UNet decoder to predict plausible object-placement regions in EO images.

Shantanu Acharya, Research Scientist at NVIDIA

March 2026 – May 2026

Independent Research Intern

- Designed **Pulsar Attention**, a distributed long-context LLM inference framework replacing static anchor duplication with content-aware statistical summaries.
- Implemented the experimental stack: distributed attention, **Max-IDF summarization**, FlashAttention-based inference, and evaluation on **RULER** and **BABILong**.
- Co-authored a paper reporting up to **4.7%** accuracy gain over dense attention, **6.1%** over Star Attention, and **3.3× lower Phase-1 FLOPs**.

Publications

More Than a Quick Glance: Overcoming the Greedy Bias in KV-Cache Compression

ICLR 2026

Preprint | 📄 GitHub | ICBINB Workshop

- Authors: **Aryan Sood**, Tanvi Sharma, Vansh Agrawal
- Developed **LASER-KV**, a KV-cache compression method achieving up to **10% higher accuracy at 128k context** over greedy pruning baselines using block-wise accumulation and **Exact-LSH sampling**.

Dropping the Anchor: Statistical Summarization for Distributed Systems via Pulsar Attention

EMNLP 2026

Preprint | 📄 GitHub | 1 EMNLP Findings | ICML AdaptFM Workshop

- Authors: **Aryan Sood**, Shantanu Acharya
- Proposed **Pulsar Attention**, enabling cross-block communication through lightweight summaries and improving long-context accuracy over dense and Star Attention baselines.

Lost in Interpolation: Why Predictive Feedback Fails in Diffusion Language Models

COLM 2026

Preprint | 📄 GitHub | Non-AR LM Workshop

- Authors: **Aryan Sood**, Ishaan Bansal, Lavanya Nigam, Vidit Aggarwal
- Introduced spherical soft-masking for diffusion LLMs, replacing Euclidean LERP feedback with **SLERP over Frechet means** to mitigate embedding norm collapse during MDLM fine-tuning.

Projects

ICLR Blogpost on Sparsity | Accepted at ICLR 2026 | Blog

Dec. 2025

- Authored a technical survey on sparse attention and efficient inference, unifying structural and data-driven sparsity while analyzing emergent attention patterns for fidelity-preserving, lower-cost inference.

Sparse Attention Pre-Selection | 📄 GitHub

Sep. 2025

- Trained a lightweight neural token-pruning model on **SQuAD attention distributions**, pruning over **60%** of context tokens and integrating TidalDecode-style sparse attention to achieve up to **1.7× decoding speedup** while retaining over **97%** of dense-attention QA performance.

Adaptive Momentum-based Decoding for Diffusion LLMs | 📄 GitHub

Dec. 2025

- Proposed a diffusion LLM decoding framework combining confidence momentum, semantic hysteresis, entropy-aware control, Elastic Sampling, and precision-monotonic scheduling, improving **GSM8K** accuracy from **~60% to ~78%** over confidence-based remasking baselines.

Achievements

- All India Rank 1519, JEE Advanced 2024** | Top exam for IIT admission, taken by 150k+ students nationally
- Within top 300 in Flipkart Gridlock** | More than 10,000 teams participated across India.
- Top 20 of 1000+ Projects** | Recognized for exceptional technical implementation among IIT Roorkee peers
- 2nd Place, PixelPlay 2025** | Week-long computer vision hackathon among top AI talent at IIT Roorkee

Extracurricular

Vision and Language Group, IIT Roorkee

Feb. 2025 – Present

Core Member | VLG Website | 📄 GitHub

IIT Roorkee

- Contribute to a deep learning research group focused on computer vision and language modeling through collaborative projects and technical discussions.